

CFD-FEM Aero-Structural Analysis of a Sounding Rocket

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Contents

1	Objectives and scope	3
2	Determination of peak aerodynamic and propulsive loads	4
2.1	Critical atmospheric conditions	4
2.2	Geometry	5
2.3	Computational domain	5
2.4	Mesh	5
2.5	Solver setup	6
2.6	Results and post-processing	6
2.7	Local mesh-independence check	7
2.8	Peak propulsive loads	7
3	Beam model design	8
3.1	Load-carrying masses	8
3.2	Non-load-carrying masses	9
4	Application of loads to the beam model	11
5	Internal resultants	12
6	Local checks of the critical components	14
6.1	Bulkhead	14
6.2	M8 screws	15
6.3	Third flange	15
6.4	M4 screws	17
6.5	Body tube	17
6.6	Electronics-bay rods	18
7	Additional analyses	20
7.1	Rail buttons	20
7.2	CubeSat support	20
8	Overview of structural verification	22
9	Appendix A: tables of concentrated and distributed masses	23

1 Objectives and scope

This report assesses the structural adequacy of a sounding rocket under a conservative combination of aerodynamic and propulsive loads. The aim is not to optimize the design, but to check whether the current configuration can safely withstand a worst-case loading condition.

The adopted method follows an aerostructural workflow. First, the external loads are estimated. These loads are then transferred to an equivalent structural model, which is used to evaluate the internal actions and to check the most critical components. For this reason, the analysis mainly focuses on the load-transfer procedure and on the resulting engineering assessment.

More specifically, the report aims to:

- define a conservative aerodynamic load case;
- combine aerodynamic and thrust loads into a single verification case;
- transfer the applied loads to an equivalent beam model;
- evaluate the resulting axial force, shear force, bending moment, and torsional moment;
- perform local structural checks on the most critical components.

The scope of the document is intentionally limited to the implementation of the CFD-to-FEM workflow and to the related structural verification. In this context, the aerodynamic analysis is used to generate the structural loads, while the structural model is used to evaluate the corresponding internal actions and the response of the main components. The report is not meant to provide a complete numerical verification of either the CFD model or the FEM model. In particular, detailed analyses of mesh convergence, domain independence, solver sensitivity, model-form uncertainty, and sensitivity to structural idealization are outside the scope of this work. These aspects are considered only where needed to support the present engineering workflow, while a broader CFD verification discussion is presented separately in the companion thesis.

The results should therefore be interpreted as part of a conservative engineering verification process, aimed at assessing the structural load path and the adequacy of the current design under the selected worst-case loading condition.

2 Determination of peak aerodynamic and propulsive loads

2.1 Critical atmospheric conditions

Aerodynamic loads are evaluated by CFD simulations in ANSYS Fluent, using the flight velocity U and the corresponding atmospheric static conditions. The critical condition is identified by recalling that [7]:

$$L = qSC_L \quad D = qSC_D \quad (1)$$

with [7]:

$$q = \frac{1}{2} \rho U^2 \quad (2)$$

Assuming a linear aerodynamic regime [7]:

$$C_L = C_{L_\alpha} \alpha \quad C_D = C_{D_0} + k C_L^2 \quad (3)$$

lift scales with $q \alpha$, while drag depends on both q and α^2 . Therefore:

$$L = qSC_{L_\alpha} \alpha \quad D = qS (C_{D_0} + k C_{L_\alpha}^2 \alpha^2) \quad (4)$$

Strictly speaking, the most critical aerodynamic condition for lift is the one that maximizes the quantity $q \alpha$, while drag is also affected by the baseline term C_{D_0} . In the present work, a conservative envelope condition is defined by combining the maximum angle of attack obtained from the trajectory analysis with the atmospheric state corresponding to the maximum dynamic pressure.

Maximum angle of attack

RocketPy trajectory analyses give the following maximum angle of attack:

$$|\alpha|_{\max} = 1.4^\circ \quad (5)$$

Maximum dynamic pressure

The dynamic pressure is maximized over the available ascent trajectory data, from lift-off to apogee. The resulting value is:

$$\max(q) = 3.774 \cdot 10^4 \text{ Pa} \quad (6)$$

The corresponding atmospheric conditions, obtained from the ISA76 model, are [10]:

$$P = 9.139 \cdot 10^4 \text{ Pa} \quad \rho = 1.127 \text{ kg/m}^3 \quad T = 282.5 \text{ K} \quad (7)$$

while the flight condition adopted in the aerodynamic simulations is:

$$U = 258.8 \text{ m/s} \quad M = 0.7680 \quad (8)$$

A second critical condition is also introduced to maximize the aerodynamic torsional moment. When $\alpha \neq 0$, a rolling moment can arise. This contribution is zero when the vehicle is symmetric with respect to the wind plane, and it reaches its maximum for a roll angle of 30° , corresponding to one fin being normal to that plane [5].

This attitude is therefore adopted in the CFD simulations, and the corresponding torsional moment is included among the target outputs.

2.2 Geometry

The CAD model of the rocket reproduces the actual external geometry, except for small local details such as screws and fasteners.

For the structural analysis, the external body was divided into several surfaces in order to extract the integrated aerodynamic loads over selected regions. The adopted subdivision is shown in Fig. 1.

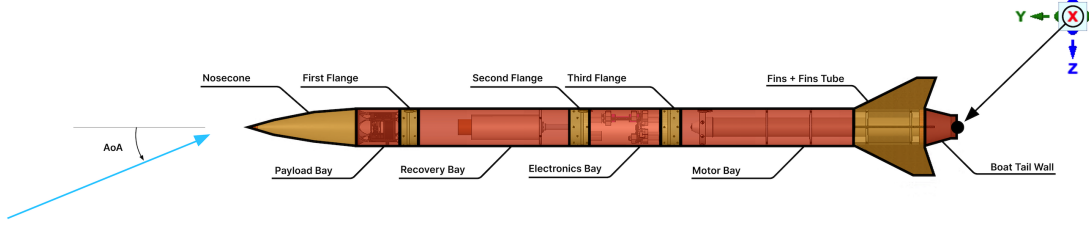


Figure 1: Surface partition adopted for aerodynamic load evaluation.

The reference frame used for load extraction is different from the one adopted in the aerodynamic chapter and is defined as follows:

- origin at the centroid of the boat-tail base;
- x -axis normal to the wind plane, positive out of the page;
- y -axis aligned with the rocket longitudinal axis, directed from the fins toward the boat-tail;
- z -axis defined so as to complete a right-handed reference system.

2.3 Computational domain

The CFD simulations were carried out in a box-shaped computational domain, whose dimensions were defined using the rocket length $l = 3$ m as reference. The domain size was chosen to reduce the effect of the outer boundaries on the aerodynamic loads. In particular:

- the inlet plane was placed $5l$ upstream of the nose cone;
- the outlet plane was placed $10l$ downstream of the base;
- the lateral far-field boundaries were placed at a distance of $0.75l$ from the fuselage.

2.4 Mesh

A global surface sizing of 4 mm was applied to the rocket geometry. The boundary-layer mesh was designed on the basis of previous CFD estimates of the maximum boundary-layer thickness, which gave a target value of about:

$$\delta_{\max} \approx 4.8 \text{ cm}$$

Assuming a target value of $y^+ = 120$, consistent with the use of y^+ -insensitive wall functions in the RANS $k-\omega$ SST framework, the resulting first-layer height is 0.46 mm. With 32 prism layers and a growth rate of 1.076, the required total thickness is reached with an additional margin of 11.8%.

A Body of Influence was also introduced around the rocket to obtain a smoother transition between the boundary-layer region and the outer tetrahedral mesh.

2.5 Solver setup

The simulations were carried out in ANSYS Fluent using a coupled pressure-based solver. The flow was treated as steady, since the characteristic flow-development times were assumed to be shorter than the characteristic times associated with changes in vehicle attitude.

Air density was computed from the ideal-gas law, while dynamic viscosity was evaluated using Sutherland’s law. Turbulence was modeled with the RANS $k-\omega$ SST model. The inlet turbulence quantities were set to the default values:

$$I = 0.05 \quad VR = 10$$

Each simulation was run for 1500 iterations, which was found to be sufficient to obtain converged aerodynamic loads.

2.6 Results and post-processing

Post-processing showed that the flow solution was consistent with the assumptions used for mesh generation. In particular, the computed boundary-layer development remained compatible with the target thickness adopted during the mesh design, and the wall resolution was coherent with the selected near-wall treatment. Over the whole rocket surface, the results satisfied:

$$y^+ < 300$$

with $y^+ \approx 120$ over most of the fuselage and only local peaks near the fins.

These results support the adequacy of the adopted mesh and solver setup for the present aerodynamic load evaluation.

For each surface element, the aerodynamic loads were averaged over the last 250 iterations. To reduce the effect of numerical noise, each load component was then filtered separately: a local contribution was retained only if it exceeded 1% of the total value of the same component obtained by summing the contributions of all surface elements. Otherwise, it was neglected.

Table 1: Aerodynamic loads integrated over the main structural elements.

Element	F_x [N]	F_y [N]	F_z [N]	M_x [N m]	M_y [N m]	M_z [N m]
Nosecone	0.04123	-21.68	-32.36	-49.70	0	-0.0593
Payload bay	0	-8.423	-1.179	-1.467	0	0
First flange	0	-3.902	0	0	0	0
Recovery bay	0	-28.59	-2.183	-1.570	0	0
Second flange	0	-3.478	0	0	0	0
Electronics bay	0.04496	-12.83	0	0	0	-0.006206
Third flange	0	-3.379	0	0	0	0
Motor bay	-0.3562	-29.35	-10.61	6.247	0	-0.2256
Fins	-3.265	-120.7	-171.4	154.4	-0.3849	-2.875
Fins tube	-1.009	-13.46	-43.26	36.61	0	-0.9602
Boat-tail	0.2499	-43.55	6.204	-5.974	0	0.3064
Total	-4.294	-289.4	-251.4	138.5	-0.3849	-3.813

The exhaust jet was not included in the present CFD model, and the nozzle exit was therefore treated as a closed surface. As a result, the pressure distribution computed in the base region below the nozzle exit includes a nonphysical contribution caused by the absence of the jet plume. Since this contribution does not represent the actual flight configuration, it was excluded from the aerodynamic loads transferred to the structural analysis. Therefore, the loads used in the structural model do not include the portion of base drag acting on the area below the nozzle exit.

2.7 Local mesh-independence check

A local mesh-sensitivity check was carried out by performing an additional simulation in which the face sizing was halved in the fin and boat-tail region. The comparison was limited to F_y , F_z , and M_x , since these are the dominant load components and therefore the most meaningful ones for evaluating local mesh sensitivity. A stricter acceptance threshold of 5% was adopted for these quantities.

The resulting variations are reported in Tab. 2.

Table 2: Local mesh-sensitivity check in the fin and boat-tail region, restricted to the dominant load components.

Load	Original sizing	Half sizing	Acceptance criterion	Measured variation
F_y [N]	-177.7	-171.5	$ \Delta_{\%} < 5\%$	$ \Delta_{\%} = 3.6\%$
F_z [N]	-208.5	-212.1	$ \Delta_{\%} < 5\%$	$ \Delta_{\%} = 1.7\%$
M_x [Nm]	185.0	187.4	$ \Delta_{\%} < 5\%$	$ \Delta_{\%} = 1.3\%$

All the selected quantities satisfy the adopted threshold, indicating that the local mesh resolution in the fin and boat-tail region is adequate for structural load extraction.

2.8 Peak propulsive loads

For the motor installed on the rocket, the Cesaroni Pro75 M1545, the maximum thrust reported in the manufacturer documentation is [12]:

$$T_{\max} = 1.840 \cdot 10^4 \text{ N} \quad (9)$$

3 Beam model design

At this stage, the internal resultants along the rocket are evaluated, namely axial force, shear force, bending moment, and torsional moment.

These quantities are obtained through structural simulations in ANSYS Static Structural using a beam idealization. The rocket is modeled as a free body, so the applied external loads are balanced by inertia forces rather than by fixed supports. This is done through the Inertia Relief option. Introducing supports would generate spurious reaction forces and moments, leading to a nonphysical evaluation of the internal resultants.

Since the structural response under inertia relief depends directly on the mass and inertia distribution, the beam model must reproduce the longitudinal distribution of mass, sectional properties, and inertia characteristics of the vehicle [3].

The adopted approach distinguishes between:

- **load-carrying masses**, i.e. structural components that transfer loads along the rocket, such as flanges, bulkheads, and aluminum tubes;
- **non-load-carrying masses**, i.e. internal subsystems not intended to carry the global structural loads, such as electronics, recovery hardware, and payload.

3.1 Load-carrying masses

Since the structural properties vary along the longitudinal direction, the beam model is divided into multiple segments, each representing a specific portion of the real vehicle and assigned its own geometric properties.

With the exception of the rods crossing the electronics bay, all load-carrying parts of the rocket are modeled as circular sections, either solid or hollow. For a generic section with outer diameter D_e and inner diameter D_i , the cross-sectional area, the transverse second moments of area, and the torsional constant are computed as [14]:

$$A = \frac{\pi}{4} (D_e^2 - D_i^2) \quad I_{xx} = I_{zz} = \frac{\pi}{64} (D_e^4 - D_i^4) \quad J = I_{xx} + I_{zz} \quad (10)$$

A separate treatment is required for the four rods crossing the electronics bay, since the actual load-carrying section is not continuous and cannot be represented directly by a single circular tube. To keep a one-dimensional beam idealization, the rod assembly is replaced by an equivalent hollow circular section with outer and inner diameters D_{e_r} and D_{i_r} .

The equivalent section is defined by requiring that it preserves:

- the total cross-sectional area of the four rods;
- the transverse bending inertias of the real rod assembly about the beam axis.

Let $d_r = 10.15$ mm be the diameter of each rod, A_r its area, and I_r its centroidal second moment of area. If the rod centers are located at a distance $r = 38.30$ mm from the beam axis, the equivalent properties are obtained from:

$$A_{eq} = 4A_r \quad I_{xx eq} = I_{zz eq} = 4(I_r + A_r r^2) \quad (11)$$

where the second relation follows from the Huygens-Steiner theorem [15].

Substituting the expressions of the equivalent hollow circular section gives:

$$\begin{cases} \frac{\pi}{4} (D_{e_r}^2 - D_{i_r}^2) = 4 \left(\frac{\pi}{4} d_r^2 \right) \\ \frac{\pi}{64} (D_{e_r}^4 - D_{i_r}^4) = 4 \left(\frac{\pi}{64} d_r^4 + \frac{\pi}{4} d_r^2 r^2 \right) \end{cases} \quad (12)$$

The two equations above are then solved for D_{e_r} and D_{i_r} , which define the equivalent tubular section adopted in the beam model.

The torsional behavior of the rod assembly is instead approximated by summing the torsional constants of the four individual rods [6], namely:

$$J_r = 4 \left(\frac{\pi}{32} d_r^4 \right) \quad (13)$$

This approximation does not reproduce the exact torsional response of the real assembly, but it provides a practical equivalent property consistent with the simplified beam model adopted here.

No equivalent beam section is introduced for the boat-tail or the nosecone. These components are instead represented as concentrated masses applied at the end nodes.

Each beam segment is assigned its corresponding material. Structural steel is used for the equivalent section representing the electronics-bay rods, while aluminum alloy is used for the remaining load-carrying members. The flanges, however, are modeled with a fictitious zero-density aluminum material. This preserves their stiffness and sectional properties without adding mass directly to the beam. Their actual mass and inertia are then reintroduced as concentrated quantities at the flange centroid, so that screws, joints, and local connections can be represented more realistically in the mass model.

The geometric properties assigned to each beam segment are summarized in Tab. 3.

Table 3: Geometric properties assigned to the beam segments representing the load-carrying structure.

Segment	y [mm]	D_e [mm]	D_i [mm]	A [mm ²]	$I_{xx} = I_{zz}$ [mm ⁴]	J [mm ⁴]
Payload bay	2313–2480	150.0	146.0	$9.299 \cdot 10^2$	$2.547 \cdot 10^6$	$5.093 \cdot 10^6$
First flange 1	2280–2313	150.0	136.0	$3.145 \cdot 10^3$	$8.058 \cdot 10^6$	$1.612 \cdot 10^7$
First flange 2	2270–2280	150.0	0.000	$1.767 \cdot 10^4$	$2.485 \cdot 10^7$	$4.970 \cdot 10^7$
First flange 3	2233–2270	150.0	136.0	$3.145 \cdot 10^3$	$8.058 \cdot 10^6$	$1.612 \cdot 10^7$
Recovery bay	1605–2233	150.0	144.0	$1.385 \cdot 10^3$	$3.744 \cdot 10^6$	$7.488 \cdot 10^6$
Second flange 1	1560–1605	150.0	136.0	$3.145 \cdot 10^3$	$8.058 \cdot 10^6$	$1.612 \cdot 10^7$
Second flange 2	1550–1560	150.0	0.000	$1.767 \cdot 10^4$	$2.485 \cdot 10^7$	$4.970 \cdot 10^7$
Second flange 3	1525–1550	150.0	136.0	$3.145 \cdot 10^3$	$8.058 \cdot 10^6$	$1.612 \cdot 10^7$
Electronics bay	1225–1525	109.5	107.6	$3.255 \cdot 10^2$	$4.791 \cdot 10^5$	$3.927 \cdot 10^3$
Third flange 1	1200–1225	150.0	136.0	$3.145 \cdot 10^3$	$8.058 \cdot 10^6$	$1.612 \cdot 10^7$
Third flange 2	1190–1200	150.0	0.000	$1.767 \cdot 10^4$	$2.485 \cdot 10^7$	$4.970 \cdot 10^7$
Third flange 3	1145–1190	150.0	136.0	$3.145 \cdot 10^3$	$8.058 \cdot 10^6$	$1.612 \cdot 10^7$
Motor bay 1	1070–1145	150.0	144.0	$1.385 \cdot 10^3$	$3.744 \cdot 10^6$	$7.488 \cdot 10^6$
Bulkhead	1045–1070	150.0	0.000	$1.767 \cdot 10^4$	$2.485 \cdot 10^7$	$4.970 \cdot 10^7$
Motor bay 2	122.6–1045	150.0	144.0	$1.385 \cdot 10^3$	$3.744 \cdot 10^6$	$7.488 \cdot 10^6$

3.2 Non-load-carrying masses

Non-load-carrying components are introduced into the beam model either as concentrated masses, including their principal moments of inertia, or as distributed masses.

In the model:

- concentrated masses are shown in red;
- distributed masses are shown as dark yellow lines;
- auxiliary zero-mass points are shown as yellow dots.

These auxiliary points are introduced where needed to reproduce the longitudinal inertia distribution more accurately, since this cannot always be represented by a single distributed mass applied along the beam axis.

A comparison between the real rocket configuration and the corresponding equivalent beam model is shown in Fig. 2.

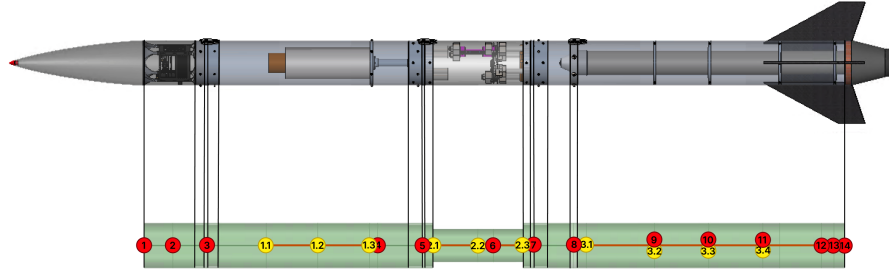


Figure 2: Equivalent beam model adopted for structural analysis.

The complete datasets adopted in the beam model are reported in Tabs. 9 and 10 of Appendix A.

4 Application of loads to the beam model

For each aerodynamic region, the CFD surface loads are replaced by an equivalent beam loading that preserves the same resultant forces and the same resultant moments about the rocket center of gravity.

The force components F_x , F_y , and F_z are applied to the corresponding beam segment as uniformly distributed loads over its span. For the nosecone and the boat-tail, whose aerodynamic loads act on the end portions of the vehicle, the corresponding resultants are applied at the beam end nodes.

This choice ensures that the beam model reproduces the correct force resultants for each aerodynamic region. However, it does not in general preserve the corresponding moment resultants. Once a distributed aerodynamic load is replaced by an equivalent beam load, its line of action is constrained by the beam idealization: for a uniformly distributed load, the resultant acts at the midpoint of the element, whereas the actual CFD center of pressure may be located elsewhere. The same issue also arises for the nosecone and boat-tail loads, since transferring them to the beam end nodes changes the corresponding moment arm.

For this reason, a corrective moment is introduced for each region so that the final beam loading is statically equivalent to the CFD loading with respect to the center of gravity. The correction is defined as:

$$\Delta \mathbf{M} = \mathbf{M}^{(\text{CFD})} - \mathbf{M}^{(\text{beam})} \quad (14)$$

where $\mathbf{M}^{(\text{beam})}$ is the moment generated by the equivalent beam-applied forces alone.

Let:

$$b = \pm \left(\frac{y_i + y_{i+1}}{2} \right) - y_{\text{CG}} \quad y_{\text{CG}} = 1160 \text{ mm} \quad (15)$$

where y_i and y_{i+1} are the endpoints of the i -th beam segment. For nosecone and boat-tail end loads, b reduces to the distance between the corresponding beam-end location and the center of gravity.

In the present beam idealization, the correction is evaluated component by component. Since F_y acts along the beam axis, it does not contribute to M_x or M_z . Moreover, the applied force resultants have zero lever arm with respect to the beam axis, so their transfer does not generate any moment about y . The corrective moments therefore reduce to:

$$\Delta M_x = M_x^{(\text{CFD})} - b F_z \quad \Delta M_y = M_y^{(\text{CFD})} \quad \Delta M_z = M_z^{(\text{CFD})} - b F_x \quad (16)$$

In practice, the most significant corrections arise in the x -direction, where the offset of the lateral force F_z with respect to the center of gravity produces non-negligible bending contributions. The correction about the y -axis is instead negligible for all regions except the fins, while the correction about the z -axis remains relatively small because the associated force component F_x is limited.

The resulting corrective moments are included in the final load set applied to the beam model, summarized in Tab. 4.

Table 4: Aerodynamic loads and corrective moments applied to the beam model.

Segment	F_x [N]	F_y [N]	F_z [N]	ΔM_x [N m]	ΔM_y [N m]	ΔM_z [N m]
Nosecone	0.04123	-21.68	-32.36	-6.985	0	-0.004876
Payload bay	0	-8.423	-1.179	-0.009167	0	0
First Flange	0	-3.902	0	0	0	0
Recovery bay	0	-28.59	-2.183	0.08690	0	0
Second Flange	0	-3.478	0	0	0	0
Electronics bay	0.04496	-12.83	0	0	0	0.003460
Third flange	0	-3.379	0	0	0	0
Motor bay	-0.3562	-29.35	-10.61	2.295	0	-0.09292
Fins	-3.265	-120.7	-171.4	4.425	-0.3849	-0.01813
Fins tube	-1.009	-13.46	-43.26	-1.243	0	-0.07733
Boat-tail	0.2499	-43.55	6.204	0.2300	0	0.05650

5 Internal resultants

The internal-resultant diagrams obtained from the beam model are shown in Fig. 3.

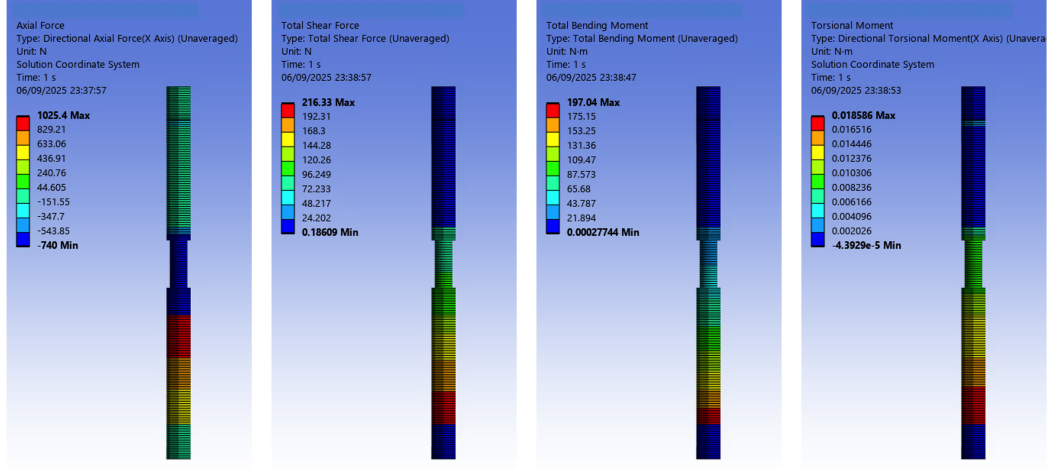


Figure 3: Internal-resultant diagrams obtained from the equivalent beam model.

In the following local analyses, each component is treated as an isolated free body. The internal resultants extracted from the beam model are therefore interpreted as the actions transmitted by the surrounding structure and applied at the upper and lower interfaces connecting the component to the rest of the rocket.

These actions are taken from the local beam solution and therefore depend on the position along the vehicle. In general, the interface actions acting at the top and bottom of a given component do not balance each other. This is consistent with the fact that, in the considered configuration, the rocket is accelerating. Static equilibrium is therefore recovered through inertia relief, with the corresponding rigid-body accelerations introduced by the solver as part of the solution.

Axial force

The axial-force diagram shows a discontinuity at the bulkhead, which corresponds to the motor-attachment region. The maximum tensile force is found below this point, while above it the structure is mainly under compression. This compressive branch extends into the third flange and the electronics bay.

For each component, the axial action is represented by two interface forces: N_a , acting from above, and N_b , acting from below. The adopted sign convention is:

$$N > 0 \text{ tension} \quad N < 0 \text{ compression} \quad (17)$$

Shear force

The maximum shear force is found in the fin region, consistently with the location of the largest lateral aerodynamic loads.

For each component, shear is represented by two transverse interface forces, T_a and T_b , acting in the beam-model plane. The adopted sign convention is:

$$T > 0 \text{ clockwise action about the component centroid} \quad T < 0 \text{ counterclockwise action} \quad (18)$$

In the following local checks, only the absolute values of the shear forces are reported.

Bending moment

The bending moment also reaches its maximum in the fin region.

For each component, bending is represented by two interface moments, M_a and M_b , extracted from the beam model in the reference plane. The adopted sign convention is:

$$M > 0 \text{ clockwise bending moment} \quad M < 0 \text{ counterclockwise bending moment} \quad (19)$$

In the local analyses, the bending load is introduced depending on the component being examined. For some parts, M_a and M_b are applied directly as moments, while for others they are converted into an equivalent axial-force distribution that varies linearly with the distance d from the neutral axis. The resulting axial stresses follow the same sign convention adopted for the axial force, namely positive in tension and negative in compression.

Torsional moment

Even in the selected worst-case torsional configuration, torsion remains small compared with the other internal actions, with the largest values again found near the fin region.

For this reason, torsion is neglected in the local verification of all components.

6 Local checks of the critical components

The local structural verifications focus on the most critical components.

The target safety factor is set to:

$$SF > 1.1 \quad (20)$$

with respect to the yield stress [13].

The representative room-temperature material properties adopted in the local structural checks are summarized in Tab. 5 [1, 11, 2, 9, 8]. When the exact material grade was not available, nominal values representative of the corresponding material family were used.

Table 5: Representative material properties adopted in the local structural checks.

Property	Al 6061-O	ASTM A36 steel	PVC-U	ABS (GP)
ρ [kg/m ³]	2700	7850	1470	1040
ν [-]	0.330	0.260	0.350	0.350
E [GPa]	68.9	200.0	3.10	2.50
G [GPa]	26.00	79.30	1.15	0.926
σ_y [MPa]	55.2	250.0	55.0	45.0
σ_u [MPa]	124.0	480.0	30.0	45.0

6.1 Bulkhead

The aluminum bulkhead is part of the thrust-transfer structure and acts as the motor-retention element, transferring the motor load to the fuselage. In this component, the axial load is much larger than shear and bending.

Applied loads

The axial loads extracted from the beam model are:

$$N_a = -740.0 \text{ N} \quad N_b = -1025 \text{ N} \quad (21)$$

The load N_a is applied at the holes connecting the bulkhead to the body tube, whereas N_b is applied over the motor-bulkhead contact surface. Since shear and bending are much smaller than the axial contribution, they are neglected in this verification.

Results

The bulkhead satisfies the target safety-factor requirement with margin, as shown in Fig. 4. The neglected shear and bending contributions are small enough not to affect this conclusion.

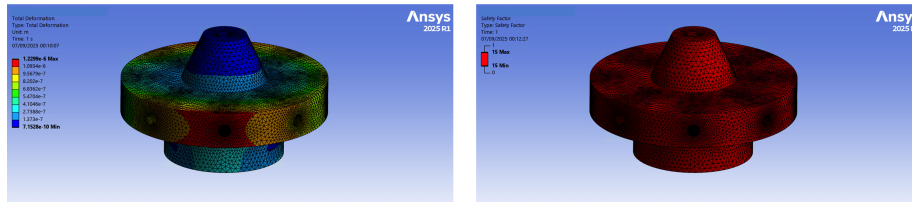


Figure 4: Deformation and safety factor distribution for the bulkhead.

6.2 M8 screws

The bulkhead is connected to the 3 mm-thick aluminum tube by seven M8 steel screws and one additional M5 screw associated with the rail button. These fasteners transfer the axial load across the bulkhead–tube interface.

Applied loads

The interface load is equally distributed among the eight fasteners, giving:

$$N_a = \frac{-740.0}{8} \text{ N} = -92.50 \text{ N} \quad N_b = \frac{-1025}{8} \text{ N} = -128.1 \text{ N} \quad (22)$$

The upper axial load acts on the shank over the 3 mm tube thickness, starting from the head side, whereas the lower axial load acts over the remaining shank length embedded in the bulkhead. No additional shear or bending loads are considered.

Results

The screw verification shows that the safety-factor requirement is satisfied, as shown in Fig. 5.

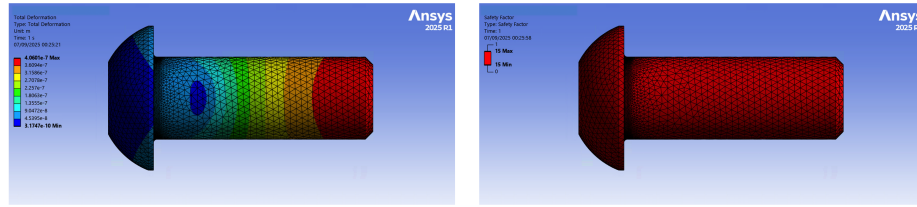


Figure 5: Deformation and safety factor distribution for the bulkhead screws.

6.3 Third flange

The aluminum flanges connect the three main structural sections of the rocket. Among them, the third flange is the most loaded one and was therefore chosen for detailed verification. Its behavior is also considered representative of the other upper flanges.

Applied loads

The loads applied to the third flange are taken from the beam model.

The axial forces at the upper and lower interfaces are approximately equal:

$$N_a \approx N_b = -740.0 \text{ N} \quad (23)$$

The upper load is applied at the holes connecting the flange to the electronics-bay rods, while the lower load is applied at the lower bolt holes.

The transverse shear forces are taken as:

$$T_a \approx T_b = 120.3 \text{ N} \quad (24)$$

These forces are applied respectively at the rod connections and at the lower holes.

The bending moments are taken as:

$$M_a \approx M_b = 87.57 \text{ N m} \quad (25)$$

This bending action is represented through equivalent axial forces applied at the connection holes according to [4]:

$$N_i = \frac{M d_i}{\sum_j d_j^2} \quad (26)$$

where d_i is the distance of the i -th hole from the neutral axis, assumed to lie at the mid-depth of the flange section by symmetry.

The equivalent axial-force distributions used to represent the bending action at the rod connections (RC) and at the lower bolt circle (LBC) are summarized below.

Table 6: Equivalent axial forces at the rod connections (RC).

d [mm]	N_i [N]
± 54	± 805.6
0	0

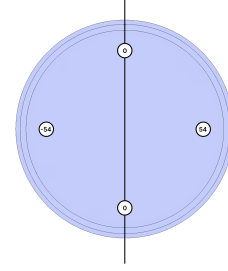


Figure 6: RC loading scheme.

Table 7: Equivalent axial forces at the lower bolt circle (LBC).

d [mm]	N_i [N]
± 70	± 205.7
± 62	± 182.2
± 34	± 100.0

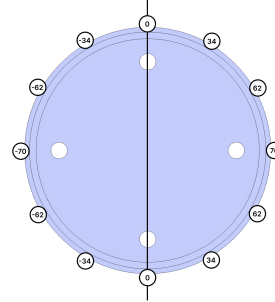


Figure 7: LBC loading scheme.

Results

The third flange withstands the applied axial, shear, and bending loads and satisfies the required safety factor, as shown in Fig. 8.

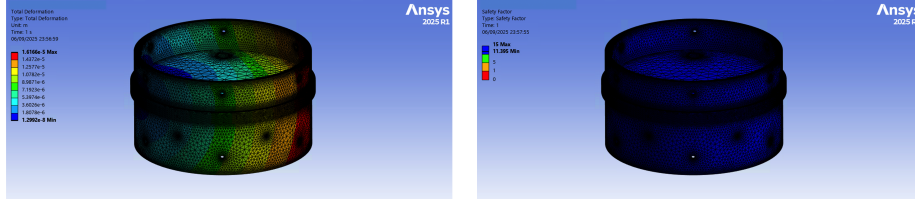


Figure 8: Deformation and safety factor distribution for the third flange.

6.4 M4 screws

The body tube is connected to the third flange by twelve M4 screws. These fasteners transfer the local axial load between the flange and the tube.

Applied loads

Since this region is mainly under compression, each screw is verified by applying one twelfth of the local axial force extracted from the beam model:

$$N_a \approx N_b = \frac{-740.0}{12} \text{ N} = -61.67 \text{ N} \quad (27)$$

As for the M8 screws, the two axial loads are applied at different axial positions, consistently with the actual attachment configuration. No additional shear or bending loads are considered.

Results

The M4 screws satisfy the target safety factor with a wide margin, as shown in Fig. 9.

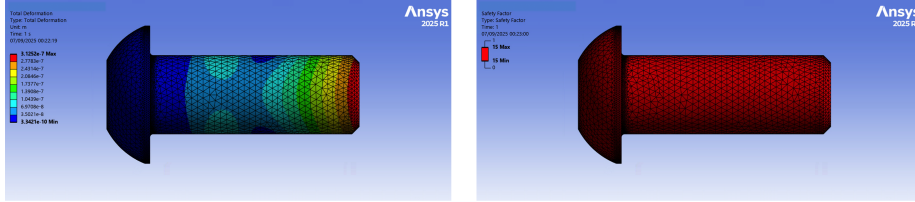


Figure 9: Deformation and safety factor distribution for the M4 screws.

6.5 Body tube

According to the beam model, the aluminum body tube reaches its maximum tensile load immediately below the bulkhead connection and its maximum compressive load immediately above it. Since this load reversal may be critical, it was checked explicitly.

Applied loads

The axial forces applied at the tube–bulkhead and tube–flange interfaces are:

$$N_a = -740 \text{ N} \quad N_b = -1025 \text{ N} \quad (28)$$

The negative sign indicates that the tube segment between the bulkhead and the flange is locally under compression.

The transverse shear force is:

$$T = 120 \text{ N} \quad (29)$$

It is applied at the flange holes, while no shear load is applied at the lower free end.

The bending action is introduced at the tube–flange bolt pattern through equivalent axial forces, using the same lower-bolt-circle distribution previously adopted for the flange.

Finally, at the lower end, where no internal load transfer takes place, the external lateral aerodynamic load generated by the fins is applied directly at the fin roots. This load is responsible for the relatively high local shear and bending in that region.

Results

The body tube satisfies the required safety factor, and the local analysis does not show any failure risk, as shown in Fig. 10.

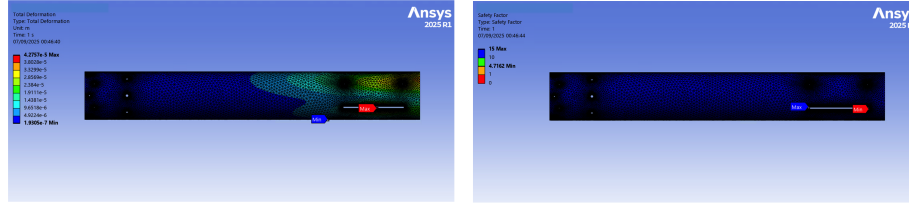


Figure 10: Deformation and safety factor distribution for the body tube.

6.6 Electronics-bay rods

The four steel rods connecting the third flange to the second flange are verified to assess their ability to transfer the local structural loads. In the real configuration, the rods are connected through prevailing-torque lock nuts, which contribute to the integrity of the assembly. In the present model, however, the lock nuts are intentionally omitted and the rods are rigidly connected to the flanges. This is a conservative assumption, made to show that the rods satisfy the required safety factor even without the beneficial effect of the actual mechanical connections.

The simulation includes both the rods and the connected flanges. Since the rods have low torsional stiffness, the torsional load is retained in the verification.

Applied loads

The axial forces applied at the upper and lower flange interfaces are:

$$N_a \approx N_b = -640 \text{ N} \quad (30)$$

The transverse shear forces applied at the same holes are:

$$T_a = 120 \text{ N} \quad T_b = 96 \text{ N} \quad (31)$$

The bending moments are applied directly at the flange holes as pure moments:

$$M_a = 65.58 \text{ N m} \quad M_b = 43.79 \text{ N m} \quad (32)$$

Unlike in the flange verification, bending is not represented here through equivalent axial forces. This simplification is acceptable because, in the present analysis, the flanges only serve to transfer the applied loads to the rods, while their detailed structural response has already been verified separately.

Results

As shown in Fig. 11, the rods satisfy the required safety factor even without the actual locking hardware. The lowest safety-factor values are found near the rod–flange connections. In the real configuration, these local effects are expected to be reduced by the presence of the lock nuts.

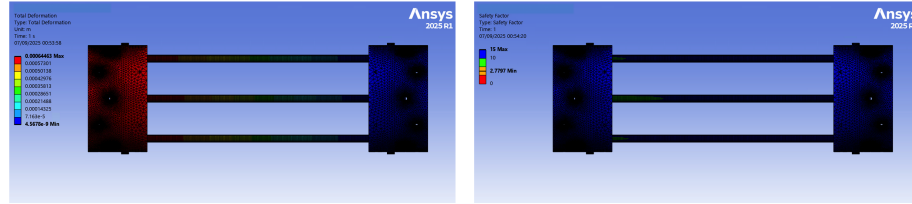


Figure 11: Deformation and safety factor distribution for the electronics-bay rods.

7 Additional analyses

7.1 Rail buttons

During launch, the rocket is supported by three PVC rail buttons connected to the fuselage by screws. Although they do not carry the global aerodynamic and thrust loads in flight, they must withstand the local reaction transmitted by the launch rail while the rocket moves along it.

Applied loads

In the present analysis, one rail button is constrained at the rail-contact point in order to represent the load path toward the launch rail.

A force equal to one third of the supported rocket weight is applied at the M5 screw hole connecting the rail button to the structure:

$$F = 88.3 \text{ N} \quad (33)$$

This value corresponds to a total rocket mass of 26.9 kg, with 9 kg attributed to the considered support point.

Results

The rail button satisfies the required safety factor, although with a limited margin, as shown in Fig. 12.

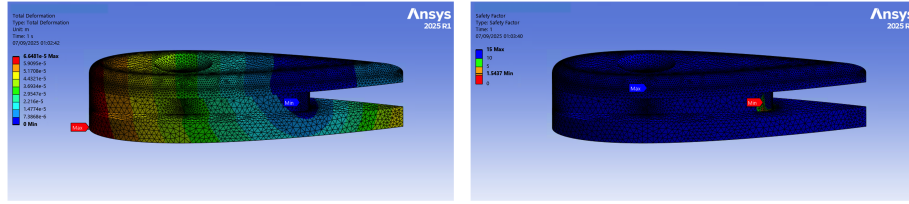


Figure 12: Deformation and safety factor distribution for the rail button.

7.2 CubeSat support

The rocket payload consists of a CubeSat supported by an ABS frame. Although this structure does not carry the global thrust and aerodynamic loads, it must withstand the inertial loads generated by the rocket acceleration.

Applied loads

The internal mass of 900 g is lumped at the four inner corner nodes of the ABS frame, corresponding to the points where the subsystem is connected to the structure. In addition, a mass of 80 g, representing the upper support, is applied at the upper external attachment points.

With inertia relief enabled, the maximum estimated rocket acceleration is taken as:

$$a = 65 \text{ m/s}^2 \quad (34)$$

Results

The support structure withstands the imposed inertial loading condition, as shown in Fig. 13.

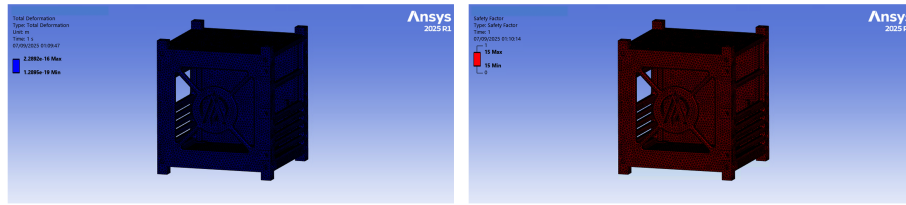


Figure 13: Deformation and safety factor distribution for the CubeSat support structure.

8 Overview of structural verification

The structural analyses carried out on the rocket were intended to verify the main design assumptions adopted throughout the analytical and numerical workflow. In particular, the goal was to assess whether the thrust-transfer structure, the motor-retention system, and the main load-bearing components can withstand the most critical combination of aerodynamic and propulsive loads expected during the mission.

The verification focused on:

- axial load transfer through the bulkhead and the flanges;
- integrity of the body tube under combined compression and bending;
- contribution of the electronics-bay rods under shear and torsion;
- adequacy of the M8 and M4 fasteners under the corresponding local loads.

The results show that all verified components satisfy the target requirement:

$$SF > 1.1 \quad (35)$$

A summary of the results is reported in Tab. 8.

Table 8: Summary of local structural verifications.

Component	Material	SF	Description
Bulkhead	Aluminum	≥ 15	Transfers the motor load to the tubular structure and is part of the main thrust-transfer system.
M8 screws	Structural steel	≥ 15	Seven M8 screws and one additional M5 screw transfer the axial load from the bulkhead to the body tube.
Third flange	Aluminum	11.40	Most loaded among the three main flanges; its verification is considered representative of the upper flanges as well.
M4 screws	Structural steel	≥ 15	Twelve M4 screws transfer the local load from the body tube to the third flange.
Body tube	Aluminum	4.716	Subjected to compression near the bulkhead and to significant bending and local lateral loading near the fin region.
Electronics-bay rods	Structural steel	2.780	Verified under a conservative assumption, neglecting the beneficial effect of the actual locking hardware.
Rail button	PVC	1.544	One of the three launch rail buttons, verified under the local support load transmitted during guided motion along the launch rail.
CubeSat support	ABS	≥ 15	Internal support frame of the CubeSat subsystem, verified under inertial loading generated by the supported internal mass.

The lowest local safety-factor values are found near the rod-flange connections in the electronics bay. However, these effects are expected to be reduced in the real configuration by the presence of the lock nuts. Likewise, torsional effects in the motor-tube and fin region remain small compared with axial and bending loads.

Overall, the analyses support the structural robustness of the thrust-transfer system and confirm the adequacy of the load-distribution strategy adopted in the beam model. The numerical results are consistent with the design assumptions and indicate that the rocket is structurally adequate for the considered mission profile.

9 Appendix A: tables of concentrated and distributed masses

Table 9: Concentrated masses: positions, axes, and inertias.

Mass	y_{GC} [mm]	Mass [g]	Axes	Inertias [$g\text{ mm}^2$]
1) Nosecone	2480	663.6	$p_1 = (0, 1, 0)$ $p_2 = (-1, 0, 0)$ $p_3 = (0, 0, 1)$	$I_{p_1} = 2.305 \cdot 10^6$ $I_{p_2} = 2.851 \cdot 10^7$ $I_{p_3} = 2.851 \cdot 10^7$
2) CubeSat	2385	516.2	$p_1 = (0, 1, 0)$ $p_2 = (-0.74, 0, -0.67)$ $p_3 = (-0.67, 0, 0.74)$	$I_{p_1} = 1.264 \cdot 10^6$ $I_{p_2} = 2.012 \cdot 10^6$ $I_{p_3} = 2.075 \cdot 10^6$
3) First flange	2273	852.9	$p_1 = (0.50, 0, 0.87)$ $p_2 = (0.87, 0, -0.50)$ $p_3 = (0, 1, 0)$	$I_{p_1} = 1.656 \cdot 10^6$ $I_{p_2} = 1.700 \cdot 10^6$ $I_{p_3} = 2.953 \cdot 10^6$
4) Peregrine system assembly	1223	664.3	$p_1 = (-0.50, 0, 0.87)$ $p_2 = (0, -1, 0)$ $p_3 = (0.87, 0, 0.50)$	$I_{p_1} = 1.357 \cdot 10^6$ $I_{p_2} = 1.614 \cdot 10^6$ $I_{p_3} = 1.816 \cdot 10^6$
5) Second flange	1560	832.5	$p_1 = (0.50, 0, 0.87)$ $p_2 = (0.87, 0, -0.50)$ $p_3 = (0, 1, 0)$	$I_{p_1} = 1.736 \cdot 10^6$ $I_{p_2} = 1.780 \cdot 10^6$ $I_{p_3} = 3.049 \cdot 10^6$
6) Electronics	1323	686.9	$p_1 = (0.03, 1, 0.01)$ $p_2 = (-1, 0, 0)$ $p_3 = (0, 0, 1)$	$I_{p_1} = 1.692 \cdot 10^6$ $I_{p_2} = 1.017 \cdot 10^7$ $I_{p_3} = 1.022 \cdot 10^7$
7) Third flange	1190	820.8	$p_1 = (0.71, 0, 0.71)$ $p_2 = (0.71, 0, -0.71)$ $p_3 = (0, 1, 0)$	$I_{p_1} = 1.735 \cdot 10^6$ $I_{p_2} = 1.735 \cdot 10^6$ $I_{p_3} = 3.010 \cdot 10^6$
8) Bulk head	1057	1110	$p_1 = (0.50, 0, 0.87)$ $p_2 = (0.87, 0, -0.50)$ $p_3 = (0, 1, 0)$	$I_{p_1} = 2.162 \cdot 10^7$ $I_{p_2} = 1.675 \cdot 10^6$ $I_{p_3} = 2.940 \cdot 10^6$
9) Centering 1	790	163.1	$p_1 = (0, 0, 1)$ $p_2 = (1, 0, 0)$ $p_3 = (0, 1, 0)$	$I_{p_1} = 3.017 \cdot 10^5$ $I_{p_2} = 3.017 \cdot 10^5$ $I_{p_3} = 6.017 \cdot 10^5$
10) Centering 2	610	163.1	$p_1 = (0, 0, 1)$ $p_2 = (1, 0, 0)$ $p_3 = (0, 1, 0)$	$I_{p_1} = 3.017 \cdot 10^5$ $I_{p_2} = 3.017 \cdot 10^5$ $I_{p_3} = 6.017 \cdot 10^5$
11) Centering 3	375	155.8	$p_1 = (0, 0, 1)$ $p_2 = (1, 0, 0)$ $p_3 = (0, 1, 0)$	$I_{p_1} = 2.869 \cdot 10^5$ $I_{p_2} = 2.869 \cdot 10^5$ $I_{p_3} = 5.721 \cdot 10^5$
12) Fins	236	418	$p_1 = (0, 0, 1)$ $p_2 = (1, 0, 0)$ $p_3 = (0, 1, 0)$	$I_{p_1} = 5.969 \cdot 10^6$ $I_{p_2} = 5.969 \cdot 10^6$ $I_{p_3} = 7.344 \cdot 10^6$
13) Centering 4	195.8	155.8	$p_1 = (0, 0, 1)$ $p_2 = (1, 0, 0)$ $p_3 = (0, 1, 0)$	$I_{p_1} = 2.869 \cdot 10^5$ $I_{p_2} = 2.869 \cdot 10^5$ $I_{p_3} = 5.721 \cdot 10^5$
14) BT + M	160	1478	$p_1 = (0, 1, 0)$ $p_2 = (0, 0, 1)$ $p_3 = (1, 0, 0)$	$I_{p_1} = 2.310 \cdot 10^6$ $I_{p_2} = 3.927 \cdot 10^6$ $I_{p_3} = 3.927 \cdot 10^6$

Table 10: Distributed and fictitious masses: spans, positions, axes, and inertias.

Mass	y_{GC} [mm]	Mass [g]	Axes	Inertias [g mm ²]
1) Recovery	1734–2076	890	$p_1 = (0, 1, 0)$	$I_{p_1} = 0$
			$p_2 = (0, 0, 1)$	$I_{p_2} = 7.748 \cdot 10^6$
			$p_3 = (1, 0, 0)$	$I_{p_3} = 7.748 \cdot 10^6$
1.1)	2076	0	$p_1 = (0, 1, 0)$	$I_{p_1} = 3.828 \cdot 10^5$
			$p_2 = (0, 0, 1)$	$I_{p_2} = 0$
			$p_3 = (1, 0, 0)$	$I_{p_3} = 0$
1.2)	1905	0	$p_1 = (0, 1, 0)$	$I_{p_1} = 3.828 \cdot 10^5$
			$p_2 = (0, 0, 1)$	$I_{p_2} = 0$
			$p_3 = (1, 0, 0)$	$I_{p_3} = 0$
1.3)	1734	0	$p_1 = (0, 1, 0)$	$I_{p_1} = 3.828 \cdot 10^5$
			$p_2 = (0, 0, 1)$	$I_{p_2} = 0$
			$p_3 = (1, 0, 0)$	$I_{p_3} = 0$
2) Plexiglass tube	1200–1550	581.3	$p_1 = (0, 1, 0)$	$I_{p_1} = 0$
			$p_2 = (-0.71, 0, 0.71)$	$I_{p_2} = 7.495 \cdot 10^6$
			$p_3 = (0.71, 0, 0.71)$	$I_{p_3} = 7.495 \cdot 10^6$
2.1)	1550	0	$p_1 = (0, 1, 0)$	$I_{p_1} = 1.047 \cdot 10^6$
			$p_2 = (-0.71, 0, 0.71)$	$I_{p_2} = 0$
			$p_3 = (0.71, 0, 0.71)$	$I_{p_3} = 0$
2.2)	1375	0	$p_1 = (0, 1, 0)$	$I_{p_1} = 1.047 \cdot 10^6$
			$p_2 = (-0.71, 0, 0.71)$	$I_{p_2} = 0$
			$p_3 = (0.71, 0, 0.71)$	$I_{p_3} = 0$
2.3)	1200	0	$p_1 = (0, 1, 0)$	$I_{p_1} = 1.047 \cdot 10^6$
			$p_2 = (-0.71, 0, 0.71)$	$I_{p_2} = 0$
			$p_3 = (0.71, 0, 0.71)$	$I_{p_3} = 0$
Motor bay	195–1015	7208	$p_1 = (0, 1, 0)$	$I_{p_1} = 0$
			$p_2 = (0, 0, 1)$	$I_{p_2} = 4.805 \cdot 10^8$
			$p_3 = (1, 0, 0)$	$I_{p_3} = 4.805 \cdot 10^8$
3.1)	1015	0	$p_1 = (0, 1, 0)$	$I_{p_1} = 1.268 \cdot 10^6$
			$p_2 = (0, 0, 1)$	$I_{p_2} = 0$
			$p_3 = (1, 0, 0)$	$I_{p_3} = 0$
3.2)	790	0	$p_1 = (0, 1, 0)$	$I_{p_1} = 1.268 \cdot 10^6$
			$p_2 = (0, 0, 1)$	$I_{p_2} = 0$
			$p_3 = (1, 0, 0)$	$I_{p_3} = 0$
3.3)	610	0	$p_1 = (0, 1, 0)$	$I_{p_1} = 1.268 \cdot 10^6$
			$p_2 = (0, 0, 1)$	$I_{p_2} = 0$
			$p_3 = (1, 0, 0)$	$I_{p_3} = 0$
3.4)	375	0	$p_1 = (0, 1, 0)$	$I_{p_1} = 1.268 \cdot 10^6$
			$p_2 = (0, 0, 1)$	$I_{p_2} = 0$
			$p_3 = (1, 0, 0)$	$I_{p_3} = 0$
3.5)	195	0	$p_1 = (0, 1, 0)$	$I_{p_1} = 1.268 \cdot 10^6$
			$p_2 = (0, 0, 1)$	$I_{p_2} = 0$
			$p_3 = (1, 0, 0)$	$I_{p_3} = 0$

Notes:

- The final concentrated mass BT+M (14) represents the boat-tail and the portion of the motor inside it.
- The transfer of the concentrated masses of the nosecone (1) and BT+M (14) was accompanied by a correction of the inertia moments using the Huygens–Steiner theorem.
- The coordinates of the principal inertia axes are expressed in the reference basis of the report.

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